

Evaluation of an Erbium Doped Waveguide Amplifier RF performance in Microwave Photonics Applications

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Abstract: Performance of an EDWA in Si₃N₄ platform is investigated for integrated MWP systems, showing stable performance under various RF conditions. The absence of induced nonlinearities confirms its suitability and reliability for integrated microwave photonic applications. © 2024 The Author(s).

1. Introduction

Over the past decade, photonic technologies have increasingly enhanced and extended the capabilities of conventional RF systems across telecommunications, defence, remote sensing, and space observation. Optical fibers provide low-loss transmission, immunity to electromagnetic interference, and broad bandwidth, making photonics particularly attractive for next-generation communications and sensing [1]. Radio-over-Fiber (RoF) technology, a key area within microwave photonics (MWP), is a prominent example, enabling the efficient transmission of RF signals over optical fiber with minimal degradation [2]. The evolution of these systems, together with the demand for compact and energy-efficient integrated solutions, has grown considerably. Photonic integrated circuits (PICs) represent a promising solution to address these requirements, allowing for integration of key components such as lasers, modulators, detectors and filters [3]. Among the essential building blocks of MWP systems, the optical amplifier plays a fundamental role in compensating for link losses, extending transmission distances and preserving signal integrity. Furthermore, erbium-doped fiber amplifiers (EDFAs) remain bulky, making them less suitable for integration in compact and lightweight photonic systems. On the other hand, semiconductor optical amplifiers (SOAs) have been widely adopted due to their high gain and straightforward integration potential. However, though easily integrated, SOAs are strongly nonlinear, generating distortions that deteriorate the RF signal quality, especially at lower frequencies [4,5]. They also exhibit high noise figure, polarization sensitivity and often require hybrid or heterogeneous integration of III-V materials on silicon or lithium niobate [6,7].

To overcome these issues, rare-earth-doped waveguide amplifiers have recently attracted renewed attention [8-10]. Among them, erbium-doped waveguide amplifiers (EDWAs) stand out due to their inherently low noise, high linearity and compatibility with low-loss photonic platforms such as silicon nitride (Si₃N₄) [11,12]. Recent demonstration achieved up to 145mW on-chip output power and a small-signal gain of more than 30 dB in a 1.2x5 mm² size [13]. In this work, we investigate the performance of a Si₃N₄ EDWA with a focus on its applicability to integrated MWP systems, analyzing frequency-dependent distortions, noise figure (NF), spurious-free dynamic range (SFDR), and demonstrate its operation in a RoF link with demanding 512-QAM modulation.

2. EDWA characterization

The EDWA under investigation consists of a 17,5 cm long Si₃N₄ spiral with a 2.5x3 mm² footprint on PIC (D147_02), as shown in Fig.1A. In the picture clearly visible is the green glowing of the waveguide when pumped due to up-conversion of Er ions. Silicon nitride spiral waveguide has 200 nm height and 5000 nm width, with three consecutive implantation steps carried out at energies of 500, 270, and 133 keV. Input and output coupling is performed with lensed optical fibers with 2.5μm MFD, resulting in 6.5 dB insertion loss per facet measured at 1630nm. EDWA is pumped by two fiber-coupled 300mW lasers at 1480 nm, one propagating alongside the signal and one counterpropagating, coupled with the signal via wavelength division multiplexing (WDM) fiber couplers. At 1550 nm, the amplifier presents an off-chip small signal gain greater than 10 dB for input powers up to -10 dBm, as reported in Fig.1B, with a measured saturated output power of 12.5 dBm. Moreover, for an input power of 0 dBm, the measured optical NF results in 6.4dB. The EDWA has also been tested at different wavelengths (Fig. 1C), showing a performance degradation, especially in gain, at 1540 nm and above 1567 nm, maintaining anyway a positive gain and a saturated output power > 9 dBm over the entire C-band, and up to 1570 nm.

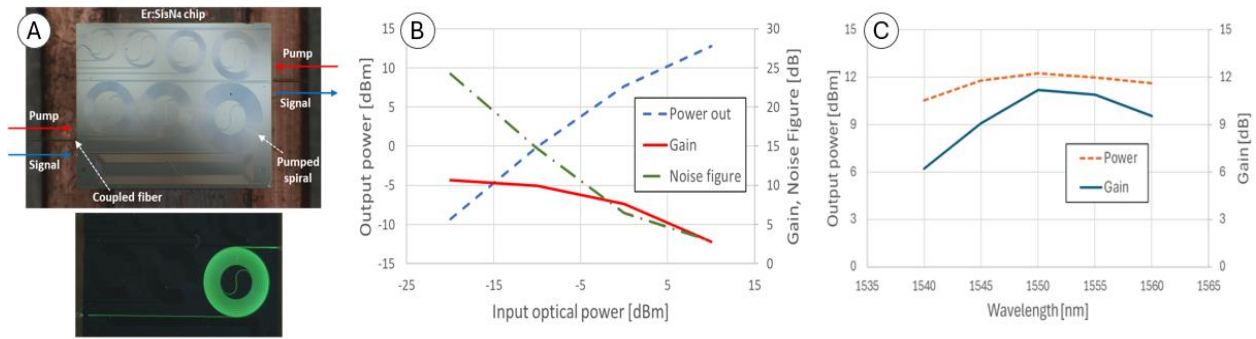


Fig.1 A) Picture of the EDWA during testing; B) Measured optical output power, Gain, and NF at different input power level; C) Gain and saturated power at different wavelengths.

3. RF characterization

To evaluate the EDWA for MWP applications, its performance in amplifying optical signals with analog modulations has been measured (Fig. 2A). A high-power laser carrier is modulated by a test RF signal via a Mach-Zehnder modulator (MZM) and coupled into the EDWA. The amplified output is detected by a wideband photodiode (PD) and measured with an electrical spectrum analyzer (ESA). A variable optical attenuator (VOA) before the EDWA adjusts its input power, and a second VOA allows to keep a constant power at the PD. The first test measured RF insertion loss across different frequencies while varying the EDWA input power. The RF signal at the MZM is set at -3 dBm, and its frequency varies from 10 MHz to 17 GHz; the optical power at PD is kept at 0 dBm. The results, reported as solid lines in Fig. 2B, show no visible effects from the EDWA, with the detected RF power only varying due to the MZM and PD frequency responses, precisely matching the Back-to-Back case with no optical amplification. On the contrary, the same measure carried out with a commercial high linearity SOA (Kamelian OPA-20-N-C) shows a very strong dependency on both RF frequency and input power, especially at lower frequencies due to the short carrier lifetime [4]. In most MWP applications linearity is fundamental, so a two tones test has been performed. In detail, the MZM is driven by two RF tones at frequencies f and $f+\Delta f$, respectively, and the third order intermodulation distortion (IMD3) at $f+2\Delta f$ is measured. Since two close optical signals could induce four wave mixing (FWM), whose efficiency increases closer the signal, the test has been repeated for Δf values of 1 kHz, 100 kHz, and 10 MHz, with f fixed at 6 GHz. The results are reported in Fig. 2C, and again show no influence of the EDWA, regardless of the input power or the frequency separation, with about 1 dB of fluctuation, well within the measurement uncertainty. Given that no nonlinear behaviors have been observed so far, the RoF link employing the EDWA is characterized in terms of NF and SFDR. Fig. 2D reports the measured NF and

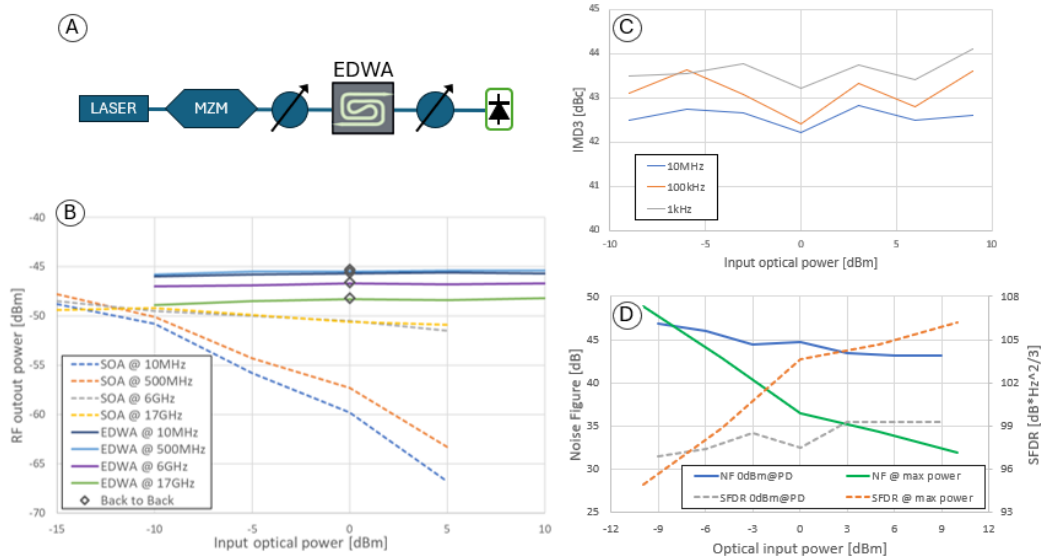


Fig. 2: A) Experimental setup; B) Detected RF power for different frequencies and input powers, with equal optical power at the photodiode; C) IMD3 to carrier levels vs EDWA optical input power, for different carriers' separation; D) NF and SFDR vs EDWA optical input power, in case of 0 dBm at the PD and without VOA.

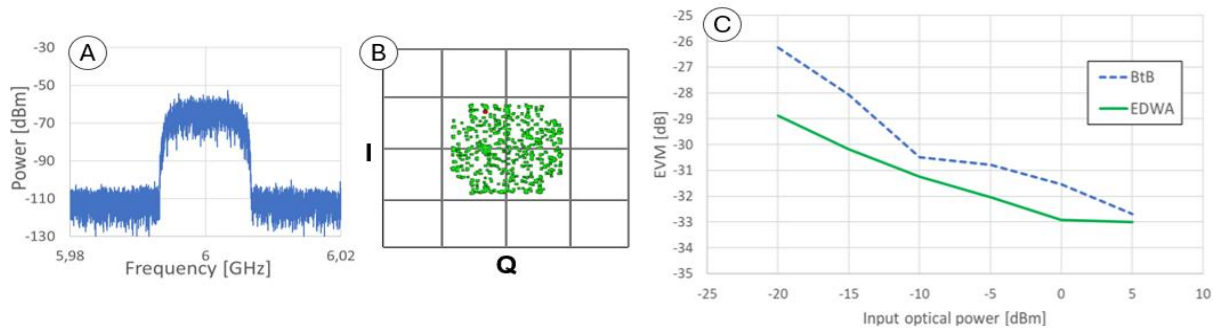


Fig. 3: A) RF spectrum of the received signal after the EDWA; B) Recovered constellation of the signal; C) EVM values measured with and without using the EDWA.

SFDR varying the input power, in case of constant optical power at the PD, and also omitting the second VOA, thus providing the PD with all EDWA output. It can be noted that, by removing the VOA, both NF and SFDR strongly improved due to the higher optical power at the PD that reduces the RF insertion loss of the link, resulting in a NF as low as 32 dB and a SFDR of 106.5 dB*Hz^{2/3} for an input power of 10 dBm with a saturated output of 11.5 dBm. Performance also improves at constant PD power, as strong EDWA saturation reduces ASE noise, lowering the electrical noise floor. Finally, as shown in Fig. 3A-B, the EDWA has been tested with complex modulations formats typically employed in RF communications, such as quadrature-amplitude modulation (QAM). In detail, the MZM is driven with a 10 MSymbols/s 512-QAM data payload at the RF frequency of 6 GHz, and the resulting error vector magnitude is calculated by a signal analyzer at the PD output (Fig. 3C). The measurement varies the optical power after the MZM, first in back-to-back (BtB) and then with EDWA amplification. The amplified signal shows improved EVM, recovering the constellation at low input power and matching BtB at saturation.

4. Conclusions

A Si₃N₄ EDWA has been presented and characterized, showing a small signal gain > 10 dB, a saturated output power of 12 dBm, and a NF of 6 dB at 0 dBm input power. Amplification of analog-modulated optical signals revealed no nonlinear behavior, confirming its suitability for MWP applications. This provides a compact alternative to bulky EDFAs and nonlinear SOAs. Although the gain is lower than commercial EDFAs, optimization of doping, waveguide length, and facet coupling (currently ~6.5 dB) could improve performance. The Si₃N₄ platform enables integration of couplers, filters, and hybrid integration with InP or LNOI, supporting fully integrated MWP systems.

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